

## PALMERSTON NORTH, New Zealand

WMO: 934040

Lat: 40.323S

Lon: 175.612E

Elev: 39

StdP: 100.86

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 0.0        | 1.2 | -1.5                          | 3.4 | 1.4  | -0.4 | 3.7 | 2.3  | 11.8                  | 11.5 | 10.6 | 11.6 | 1.2                   | 110  | 0.553 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 9.3                    | 25.9            | 18.7 | 24.7 | 18.2 | 23.5 | 17.6 | 20.2                | 23.8 | 19.3 | 22.8 | 18.5 | 22.0 | 5.1                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.9                            | 13.8 | 21.8 | 18.0 | 13.0 | 21.2 | 17.1 | 12.3 | 20.6 | 58.1          | 23.8 | 55.3 | 22.7 | 52.6 | 22.1 | 23.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |          |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|----------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |          |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max      | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)      |    |    |
| (4) 11.5          | 10.2 | 9.2 | -2.2                       | 28.7 | 0.7                | 1.7 | -2.7                                               | 30.0 | -3.1       | 31.0 | -3.5       | 31.9 | -4.1       | 33.2 (4) |    |    |
| (5)               |      |     | -2.6                       | 21.7 | 0.7                | 0.9 | -3.1                                               | 22.3 | -3.5       | 22.9 | -3.9       | 23.4 | -4.4       | 24.0 (5) |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 13.2   | 17.8  | 18.2  | 16.6  | 13.9  | 11.6  | 9.2   | 8.8   | 9.4   | 10.9  | 12.2  | 13.8  | 16.3 |
|                                                                                  | DBStd       | 4.08   | 2.57  | 2.36  | 2.49  | 2.64  | 2.71  | 2.70  | 2.47  | 2.24  | 2.24  | 2.26  | 2.33  | 2.46 |
|                                                                                  | HDD10.0     | 184    | 0     | 0     | 0     | 3     | 16    | 47    | 53    | 38    | 17    | 8     | 2     | 0    |
|                                                                                  | HDD18.3     | 1946   | 41    | 28    | 65    | 134   | 209   | 273   | 297   | 278   | 224   | 189   | 138   | 71   |
|                                                                                  | CDD10.0     | 1349   | 242   | 230   | 203   | 120   | 64    | 24    | 15    | 19    | 42    | 77    | 115   | 197  |
|                                                                                  | CDD18.3     | 69     | 24    | 24    | 10    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 9    |
|                                                                                  | CDH23.3     | 294    | 114   | 104   | 39    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 32   |
| CDH26.7                                                                          | 19          | 10     | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     |      |
| Wind                                                                             | WSAvg       | 4.2    | 4.5   | 4.3   | 4.1   | 3.7   | 3.7   | 3.6   | 3.8   | 3.8   | 4.4   | 4.7   | 4.9   | 4.4  |
| Precipitation                                                                    | PrecAvg     | 1146   | 75    | 75    | 76    | 85    | 98    | 113   | 108   | 103   | 104   | 112   | 96    | 101  |
|                                                                                  | PrecMax     | 1521   | 145   | 343   | 188   | 231   | 215   | 250   | 181   | 192   | 283   | 234   | 184   | 197  |
|                                                                                  | PrecMin     | 908    | 30    | 14    | 25    | 27    | 30    | 55    | 22    | 31    | 17    | 44    | 28    | 49   |
|                                                                                  | PrecStd     | 153    | 28    | 70    | 36    | 48    | 44    | 41    | 42    | 35    | 52    | 45    | 42    | 37   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 28.0  | 27.5  | 26.2  | 22.9  | 19.8  | 16.7  | 16.0  | 16.4  | 18.8  | 20.5  | 23.7  | 26.0 |
|                                                                                  |             | MCWB   | 19.8  | 19.1  | 18.2  | 17.4  | 15.7  | 13.7  | 12.8  | 12.6  | 14.4  | 15.3  | 17.7  | 18.6 |
|                                                                                  | 2%          | DB     | 25.9  | 25.9  | 24.3  | 21.0  | 18.1  | 15.1  | 14.4  | 15.1  | 17.1  | 18.8  | 21.4  | 24.1 |
|                                                                                  |             | MCWB   | 18.9  | 18.7  | 17.6  | 16.4  | 15.1  | 12.8  | 11.8  | 11.8  | 13.4  | 14.5  | 16.4  | 17.8 |
|                                                                                  | 5%          | DB     | 24.5  | 24.7  | 22.9  | 19.7  | 17.0  | 14.2  | 13.6  | 14.1  | 16.0  | 17.6  | 19.7  | 22.7 |
|                                                                                  |             | MCWB   | 18.2  | 18.3  | 17.1  | 15.6  | 14.3  | 12.2  | 11.2  | 11.2  | 12.9  | 13.8  | 15.1  | 17.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 10%         | DB     | 23.0  | 23.2  | 21.5  | 18.5  | 15.9  | 13.4  | 12.6  | 13.2  | 14.9  | 16.4  | 18.3  | 21.2 |
|                                                                                  |             | MCWB   | 17.5  | 17.6  | 16.5  | 15.1  | 13.7  | 11.6  | 10.5  | 10.7  | 12.2  | 13.2  | 14.4  | 16.6 |
|                                                                                  | 0.4%        | WB     | 21.5  | 21.5  | 20.0  | 18.5  | 16.7  | 14.5  | 13.6  | 13.4  | 15.2  | 16.3  | 18.6  | 20.2 |
|                                                                                  |             | MCDB   | 26.3  | 24.6  | 23.7  | 21.2  | 18.8  | 15.5  | 15.2  | 15.4  | 17.6  | 19.2  | 22.3  | 23.5 |
|                                                                                  | 2%          | WB     | 20.2  | 20.2  | 18.7  | 17.2  | 15.7  | 13.4  | 12.2  | 12.4  | 14.1  | 15.1  | 17.0  | 19.0 |
|                                                                                  |             | MCDB   | 24.1  | 23.6  | 22.4  | 19.6  | 17.4  | 14.7  | 14.0  | 14.3  | 16.2  | 17.6  | 20.3  | 22.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | WB     | 19.2  | 19.4  | 17.8  | 16.4  | 14.8  | 12.5  | 11.4  | 11.6  | 13.4  | 14.3  | 16.0  | 18.0 |
|                                                                                  |             | MCDB   | 22.8  | 22.7  | 21.5  | 18.8  | 16.5  | 14.0  | 13.1  | 13.5  | 15.4  | 16.7  | 18.6  | 21.2 |
|                                                                                  | 5% WB       | WB     | 18.2  | 18.5  | 17.1  | 15.5  | 13.8  | 11.7  | 10.7  | 11.0  | 12.6  | 13.6  | 15.0  | 17.2 |
|                                                                                  |             | MCDB   | 21.8  | 21.9  | 20.6  | 17.9  | 15.6  | 13.2  | 12.3  | 12.8  | 14.6  | 16.0  | 17.6  | 20.3 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.345  | 0.335 | 0.318 | 0.306 | 0.295 | 0.291 | 0.287 | 0.299 | 0.324 | 0.333 | 0.329 | 0.345 |      |
|                                                                                  | taud        | 2.493  | 2.546 | 2.588 | 2.612 | 2.615 | 2.600 | 2.613 | 2.581 | 2.487 | 2.474 | 2.504 | 2.492 |      |
|                                                                                  | Ebn at Noon | 984    | 969   | 944   | 891   | 837   | 805   | 838   | 885   | 921   | 960   | 994   | 989   |      |
|                                                                                  | Edh at Noon | 113    | 102   | 89    | 75    | 64    | 60    | 63    | 76    | 97    | 108   | 111   | 114   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 6.43   | 5.62  | 4.41  | 3.00  | 2.00  | 1.52  | 1.77  | 2.51  | 3.57  | 4.68  | 5.80  | 6.10  |      |
|                                                                                  | RadStd      | 0.43   | 0.54  | 0.26  | 0.21  | 0.11  | 0.10  | 0.14  | 0.16  | 0.25  | 0.29  | 0.51  | 0.45  |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |          |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|----------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3  |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)      |
| (46) Station Only           |  | +0.30 | N/A     | +0.45  | N/A     | N/A   | N/A   | -39         | -124    | +105    | N/A (46) |
| (47) Regional (8 neighbors) |  | N/A   | +0.32   | +0.55  | N/A     | N/A   | N/A   | -24         | N/A     | N/A     | N/A (47) |

Nomenclature: See separate page